# Burak Varıcı

*Curriculum Vitae*

*Machine Learning Department*
*Carnegie Mellon University*
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*bvarici.github.io*

**Research Interests:** I am broadly interested in the theoretical foundations of AI/ML, particularly in representation learning and causality. This involves developing novel theoretical results and scalable algorithms for identifiable representation learning, causal discovery, and intervention design, with a recent focus on applications in robotics, gene regulatory networks, and inverse problems.

## Academic Positions

07.2026 – present **University of Delaware**, Newark, DE.
Assistant Professor, Electrical and Computer Engineering Department
Resident Faculty of the Data Science Institute

07.2024 – 06.2026 **Carnegie Mellon University**, Pittsburgh, PA.
- Postdoctoral Research Associate, Machine Learning Department
- Host: Prof. Pradeep Ravikumar

## Education

2024 **Rensselaer Polytechnic Institute**, Troy, NY.
- Ph.D. in Electrical Computer, and Systems Engineering
- Advisor: Prof. Ali Tajer
- Dissertation: Causal Learning via Interventions: Estimation and Design

2020 **Rensselaer Polytechnic Institute**, Troy, NY.
- M.S. in Electrical Computer, and Systems Engineering

2018 **Bogazici University**, Istanbul, Turkey.
- B.S. in Electrical and Electronics Engineering

## Professional Experience

09–12.2022 **MIT-IBM Watson AI Lab**, Cambridge, MA.
- Visiting Research Scholar
- Collaborators: Dr. Dmitriy A Katz, Dr. Prasanna Sattigeri, Dr. Dennis Wei

05–08.2020 **IBM Research**, remote.
- Research Intern
- Mentors: Dr. Karthikeyan Shanmugam, Dr. Prasanna Sattigeri
- Project: Disentangled generative modeling

06–08.2017 **Speech Enabled Smart Technologies**, Istanbul, Turkey.
- Research Intern
- Project: Built neural networks for speaker identity verification.

## Publications

** denotes equal contribution .*

### Peer-reviewed Journals

JMLR'25 Score-based Causal Representation Learning: Linear and General Transformations,
**B. Varıcı**, E. Acartürk, K. Shanmugam, A. Kumar, and A. Tajer.
*Journal of Machine Learning Research, 2025*

TMLR'24 Separability Analysis for Causal Discovery in Mixture of DAGs,
**B. Varıcı**, D. Katz-Rogozhnikov, D. Wei, P. Sattigeri, and A. Tajer.
*Transactions on Machine Learning Research, 2024*

JSAIT'24 Robust Causal Bandits for Linear Models,
Z. Yan, A. Mukherjee, **B. Varıcı**, and A. Tajer.
*IEEE Journal on Selected Areas in Information Theory, 2024*

JMLR'23 Causal Bandits for Linear Structural Equation Models,
**B. Varıcı**, K. Shanmugam, P. Sattigeri, and A. Tajer.
*Journal of Machine Learning Research, 2023*

## Peer-reviewed Conferences

*AI/ML conferences are peer-reviewed with low acceptance rates (20-30%) .*

NeurIPS'26 Beyond Eigenfunctions: Divergence Principal Functions for Representation Learning,
R. Ray, S. Dharod, **B. Varıcı**, N. Boffi, and P. Ravikumar.
*Conference on Neural Information Processing Systems, 2026*

ICML'25 Contextures: Representations from Contexts,
R. Zhai, K. Yang, **B. Varıcı**, CP. Tsai, and P. Ravikumar.
*International Conference on Machine Learning, 2025*

AISTATS'25 On the Consistent Recovery of Joint Distributions from Conditionals,
M. Majid, R. Pukdee, V. Agrawal, **B. Varıcı**, and P. Ravikumar.
*International Conference on Artificial Intelligence and Statistics, 2025*

NeurIPS'24 Linear Causal Representation Learning from Unknown Multi-node Interventions,
**B. Varıcı**, E. Acartürk, K. Shanmugam, and A. Tajer.
*Conference on Neural Information Processing Systems, 2024*

NeurIPS'24 Sample Complexity of Interventional Causal Representation Learning,
E. Acartürk, **B. Varıcı**, K. Shanmugam, and A. Tajer.
*Conference on Neural Information Processing Sytems, 2024*

NeurIPS'24 Interventional Causal Discovery in a Mixture of DAGs,
**B. Varıcı**, D. Katz-Rogozhnikov, D. Wei, P. Sattigeri, and A. Tajer.
*Conference on Neural Information Processing Systems, 2024*

AISTATS'24 (oral) General Identifiability and Achievability for Causal Representation Learning,
**B. Varıcı**, E. Acartürk, K. Shanmugam, and A. Tajer.
*International Conference on Artificial Intelligence and Statistics, May 2024*

ISIT'24 Improved Bound for Robust Causal Bandits with Linear Models,
Z. Yan, A. Mukherjee, **B. Varıcı**, and A. Tajer.
*International Symposium on Information Theory, 2024*

UAI'22 Intervention Target Estimation in the Presence of Latent Variables,
**B. Varıcı**, K. Shanmugam, P. Sattigeri, and A. Tajer.
*The Conference on Uncertainty in Artificial Intelligence, 2022*

NeurIPS'21 Scalable Intervention Target Estimation in Linear Models,
**B. Varıcı**, K. Shanmugam, P. Sattigeri, and A. Tajer.
*Conference on Neural Information Processing Systems, 2021*

AISTATS'21 Learning Shared Subgraphs in Ising Model Pairs,
**B. Varıcı**$^*$, S. Sihag$^*$, and A. Tajer.
*International Conference on Artificial Intelligence and Statistics, 2021*

## Preprints and Others

2026 Block Disentanglement in CRL: Bridging Identifiability and Visual State Estimation ,
E. Acartürk, P. Kulkarni, P.Datta, K. Shanmugam, **B. Varıcı**, and A. Tajer.
arXiv:2610.06809, 2026

2026 Learning Generative Factors via Causal Representation Learning: A Score-based Perspective,
E. Acartürk, **B. Varıcı**, K. Shanmugam, and A. Tajer.
IEEE BITS the Information Theory Magazine

2026 Tabular Numeric Stretch Transformation ,
Z. Ye, J. Kim, J. Sundberg, **B. Varıcı**, and P. Ravikumar.
arXiv:2608.09162, 2026

2025 Eigenfunction Extraction for Ordered Representation Learning,
**B. Varıcı***, C.P. Tsai*, R. Ray, N. Boffi, and P. Ravikumar.
*arXiv:2510.24672*

2025 ROPES: Robotic Pose Estimation via Score-based Causal Representation Learning,
P. Kulkarni, P. Datta, E. Acartürk, **B. Varıcı**, K. Shanmugam, and A. Tajer.
*arXiv:2510.20884, NeurIPS 2025 Workshop on Embodied World Models*

2023 Score-based Causal Representation Learning with Interventions,
**B. Varıcı**, E. Acartürk, K. Shanmugam, A. Kumar, and A. Tajer.
*arXiv:2301.08230, 2023*

### Book Chapters

2025 Causal Representation Learning,
**B. Varıcı**, C. Squires, and P. Ravikumar.
Chapter 10 of Neuro-symbolic AI: Foundations and Applications, Wiley, April 2026.

## Honors & Awards

2025 IEEE Signal Processing Society Best PhD Dissertation Award
2025 NeurIPS Top Reviewer
2020–2024 IBM AI Horizons Fellowship
2024 Allen B. Dumont Prize (RPI ECSE doctoral dissertation award)
2023 NeurIPS Top Reviewer, UAI Top Reviewer
2022 Belsky Award for RPI Computational Sciences and Engineering
2022 Jerry Dziuba ECSE Graduate Student Service Award
2013–2018 TUBITAK (Turkish Equivalent of NSF) Undergraduate Fellowship
2012 Silver Medal, Turkish National Mathematical Olympiad
2012 Silver Medal, International Balkan Mathematical Olympiad
2010 Gold Medal, International Junior Balkan Mathematical Olympiad

## Talks

**Identifiable Representation Learning**
01.2026 INFORMED-AI Seminar Series - University of Bristol
**Causal Representation Learning**
10.2026 BIRS Identifiable Representation Learning Workshop
10.2025 CMU - Statistics and Machine Learning Reading Group
10.2025 KUIS (Koc University & Is Bank) AI Center
03.2025 Artificial Intelligence with Causal Techniques Workshop at AAAI
02.2025 Booth School of Business at University of Chicago – Aragam Group
03.2024 CMU - Statistical & Symbolic Learning Group
12.2023 NeurIPS Causal Representation Learning Workshop
**Causal Bandits**
02.2023 IBM Causal Reinforcement Learning Group

## Teaching

Fall 2026 **Instructor**, University of Delaware.
- ELEG 467/667: Topics in Causality and Machine Learning
  Developed and taught a new course on foundations of modern causal inference and connections to machine learning, including an introduction to areas of active research such as representation learning.

2024–2025 **Guest Lecturer**, Carnegie Mellon University.
- CMU 10716: Advanced Machine Learning - Clustering
- CMU 10716: Advanced Machine Learning - Causality
- CMU 10741: Representation Learning - Causal Representation Learning

2018–2020 **Teaching Assistant**, Rensselaer Polytechnic Institute.
- ECSE 2410: Signals and Systems
- ECSE 2610: Computer Components and Operations
- ECSE 1010: Introduction to Electrical, Component, and Systems Engineering

## Mentorship

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| 2025–2026 | Utkarsh Priyam (CMU M.S. → CMU Ph.D.) |
| 2025–2026 | Zihao Ye (CMU M.S. → Woodwide AI) |
| 2025 | Arnav Mantro (CMU Undergrad → Bloomberg Data Science) |
| 2025 | Hugo Contant (CMU Undergrad) |
| 2024 | Vishwajeet Agrawal (CMU M.S. → Skild AI) |
| 2024 | Mahbod Majod (CMU Ph.D. → MIT Ph.D.) |
| 2022-2024 | Emre Acartürk (RPI Ph.D.) |
| 2022-2024 | Zirui Yan (RPI Ph.D.) |

## Academic Service

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|---|---|
| Organizer | Co-organizer of **Identifiable Representation Learning Workshop** in October 2026, The Banff International Research Station, Banff, Alberta, Canada |
| Tutorial | Co-lead presenter of Causal Representation Learning Tutorial, AAAI 2025, Philadelphia, PA |
| Tutorial | (Contributor) Causal Representation Learning at NASIT 2025 and ICASSP 2026. |
| Journal | Reviewer for IEEE Transactions on Signal Processing, IEEE Transactions on Pattern Recognition and Machine Intelligence, Transactions on Machine Learning Research |
| Conference | Regularly reviewing for conferences NeurIPS, ICML, AISTATS, AAAI, UAI, CLeaR since 2021 |
| Committee Member | Secretary at RPI ECSE Graduate Student Council (2019–2022) |